

DATABASE DESIGN & IMPLEMENTATION DOCUMENT

NAILVERSE

Not just NAILS.. IT's the WHOLE MOOD
Online Nail Art & Nail Beauty Online Platform

WEEK – 3

3. Supplier and Inventory Management System

Scenario

Nailverse sources nail-care products — gel polishes, acrylic powders, extension tips, and cuticle oils — from multiple suppliers so technicians always have what they need for a booked service. The system should maintain supplier details, track which products each supplier provides, and monitor inventory levels to ensure product availability before an appointment is confirmed.

1. Create Supplier, Product and Inventory Tables

```
SQL> CREATE TABLE Supplier(Supplier_ID VARCHAR2(7), Supplier_Name VARCHAR2(25),Email  
VARCHAR2(30), Phone NUMBER(10), Address VARCHAR2(45));
```

Table created.

```
SQL> CREATE TABLE Product(Product_ID VARCHAR2(7),  
2 Product_Name VARCHAR2(30),  
3 Product_Type VARCHAR2(20));
```

Table created.

```
SQL> CREATE TABLE Inventory(Inventory_ID VARCHAR2(7),  
2 Product_ID VARCHAR2(7), Supplier_ID VARCHAR2(7),  
3 Stock_Quantity NUMBER, Stock_Status VARCHAR2(11),  
4 Last_Updated DATE);
```

Table created.

2. Establish Relationships

Supplier and Product

One supplier can supply multiple products.

One product can be supplied by one or multiple suppliers.

Relationship:

Supplier → Product is one to many

Product and Inventory

Each product has inventory information.

Inventory tracks available stock.

Relationship:

Product → Inventory is one to one

Primary Key added for all three tables:

```
SQL> ALTER TABLE Supplier ADD CONSTRAINT PK_Supplier PRIMARY KEY(Supplier_ID);
Table altered.

SQL> ALTER TABLE Product ADD CONSTRAINT PK_Product PRIMARY KEY(Product_ID);
Table altered.

SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);
Table altered.
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Inventory ADD
  2  CONSTRAINT FK_Prod FOREIGN KEY(Product_ID) REFERENCES Product(Product_ID);
Table altered.

SQL> ALTER TABLE Inventory ADD
  2  CONSTRAINT FK_Supplier FOREIGN KEY(Supplier_ID) REFERENCES Supplier(Supplier_ID);
Table altered.
```

Constraints to make Unique Email ID:

```
SQL> ALTER TABLE Supplier ADD
  2  CONSTRAINT Un_mail UNIQUE(Email);
Table altered.
```

To make Email NOT NULL

```
SQL> ALTER TABLE Supplier MODIFY Email NOT NULL;
Table altered.
```

Add Constraint to Check the Stock Quantity is Positive

```
SQL> ALTER TABLE Inventory ADD
  2  CONSTRAINT CK_Quan CHECK(Stock_Quantity >=0);
Table altered.
```

3. Insert Values

Insert Values to Supplier Table

```
SQL> INSERT INTO Supplier VALUES('&SupID','&SupplierName','&email',&phone,'&Address')
  2 ;
Enter value for supid: S00001
Enter value for suppliername: GlamNail Supplies
Enter value for email: glamnailsupplies@gmail.com
Enter value for phone: 8765432109
Enter value for address: 1, Anna Nagar, Chennai
```

```
old 1: INSERT INTO Supplier VALUES('&SupID','&SupplierName','&email','&phone','&Address')
new 1: INSERT INTO Supplier VALUES('S00001','GlamNail
Supplies','glamnailsupplies@gmail.com',8765432109,'1, Anna Nagar, Chennai')
```

1 row created.

```
SQL> SELECT * FROM Supplier;
```

SUPPLIE	SUPPLIER_NAME	EMAIL	PHONE

ADDRESS			

S00001	GlamNail Supplies 1, Anna Nagar, Chennai	glamnailsupplies@gmail.com	8765432109
S00002	Polish Point Co. 62, Thirumal Nagar, Thanjavur	polishpointco@gmail.com	9876543012
S00003	ArtNail Traders 90, George Town, Chennai	artnailtraders@gmail.com	7890654312

SUPPLIE	SUPPLIER_NAME	EMAIL	PHONE

ADDRESS			

S00004	ShineCraft Distributors 23, Thillai Nagar, Trichy	shinecraftdist@gmail.com	7824003009

Insert Values to Product Table

```
SQL> INSERT INTO Product VALUES('&ProdID','&ProductName','&ProdType');
Enter value for prodid: P100001
Enter value for productname: Gel Polish - Ruby Red
Enter value for prodtype: Polish
old 1: INSERT INTO Product VALUES('&ProdID','&ProductName','&ProdType')
new 1: INSERT INTO Product VALUES('P100001','Gel Polish - Ruby Red','Polish')
```

1 row created.

```
SQL> SELECT * FROM Product;
```

PRODUCT	PRODUCT_NAME	PRODUCT_TYPE

P100001	Gel Polish - Ruby Red	Polish
P100002	Acrylic Powder - Clear	Acrylic
P200001	Nail Extension Tips - Almond	Extension
P200002	Cuticle Oil - Lavender	Care

Insert Values to Inventory Table

```
SQL> INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sup',
2 &Stock,'&Stock_status',TO_DATE('&Last_upd','YYYY-MM-DD'));
Enter value for invid: Inv0001
Enter value for prodid: P100001
Enter value for sup: S00001
old 1: INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sup',
new 1: INSERT INTO Inventory VALUES('Inv0001','P100001','S00001',
Enter value for stock: 4
Enter value for stock_status: Available
Enter value for last_upd: 2026-09-12
old 2: &Stock,'&Stock_status',TO_DATE('&Last_upd','YYYY-MM-DD'))
```

```
new 2: 4, 'Available', TO_DATE('2026-09-12', 'YYYY-MM-DD'))

1 row created.

SQL> SELECT * FROM Inventory;

INVENTO PRODUCT SUPPLIE STOCK_QUANTITY STOCK_STATU LAST_UPDA
-----
Inv0001 P100001 S00001          4 Available  12-SEP-26
Inv0002 P100001 S00002          3 Available  13-AUG-26
Inv0003 P100002 S00003          4 Unavailable 15-SEP-26
Inv0004 P200002 S00004          3 Available  14-SEP-26
```

Report 1: Supplier-wise Product Report

Display:

- Supplier name
- Number of products supplied
- Available stock

```
SQL> SELECT s.Supplier_Name,
2 COUNT(i.Product_ID) AS Number_of_Products,
3 SUM(i.Stock_Quantity) AS Available_Stock
4 FROM Supplier s
5 JOIN Inventory i ON s.Supplier_ID = i.Supplier_ID
6 WHERE i.Stock_Status = 'Available'
7 GROUP BY s.Supplier_Name;

SUPPLIER_NAME          NUMBER_OF_PRODUCTS AVAILABLE_STOCK
-----
GlamNail Supplies          1              4
Polish Point Co.           1              3
ShineCraft Distributors    1              3
```

Use Joins to get details from Supplier and Inventory Table.

Report 2: Stock Availability Report

Display:

- Product name
- Stock quantity
- Stock status

```
SQL> SELECT p.Product_Name, i.Stock_Quantity, i.Stock_Status
2 FROM Product p
3 JOIN Inventory i ON p.Product_ID = i.Product_ID;

PRODUCT_NAME          STOCK_QUANTITY STOCK_STATU
-----
Gel Polish - Ruby Red          4 Available
Gel Polish - Ruby Red          3 Available
Acrylic Powder - Clear        4 Unavailable
Cuticle Oil - Lavender        3 Available
```

Report 3: Inventory Analysis

Find:

- Total products available

```
SQL> SELECT COUNT(*) AS Total_Available_Products
2 FROM Inventory
```

```

3 WHERE Stock_Status = 'Available';

TOTAL_AVAILABLE_PRODUCTS
-----
3

```

- Products out of stock

```

SQL> SELECT COUNT(*) AS OutOfStock_Products
2 FROM Inventory
3 WHERE Stock_Status = 'Out of Stock';

OUTOFSTOCK_PRODUCTS
-----
0

```

- Highest stocked products

```

SQL> SELECT p.Product_Name, i.Stock_Quantity
2 FROM Product p
3 JOIN Inventory i ON p.Product_ID = i.Product_ID
4 WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);

PRODUCT_NAME                STOCK_QUANTITY
-----
Gel Polish - Ruby Red                4
Acrylic Powder - Clear                4

```

- Average inventory quantity

```

SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory
2 FROM Inventory;

AVERAGE_INVENTORY
-----
3.5

```